

MA201A – Lecture 01 Practice & Homework

Stewart 9e §12.5 & §12.6 · Fully Worked Solutions & Active-Recall Worksheet

1 · Fully Worked Exam & Homework Problems

🔗 P1 · Line through Two Points: $P_0(-1, -4, -4)$ and $P_1(2, 0, -3)$

1. Direction Vector:

$$\vec{v} = \overrightarrow{P_0P_1} = \langle 2 - (-1), 0 - (-4), -3 - (-4) \rangle = \langle 3, 4, 1 \rangle$$

2. Parametric Equations (base P_0 , $t \in \mathbb{R}$):

$$x(t) = -1 + 3t, \quad y(t) = -4 + 4t, \quad z(t) = -4 + t$$

3. Symmetric Equations (all direction numbers $a, b, c \neq 0$):

$$\frac{x+1}{3} = \frac{y+4}{4} = \frac{z+4}{1}$$

4. Segment Equation:

$$\vec{r}(t) = (1-t)\vec{r}_0 + t\vec{r}_1 = \langle -1 + 3t, -4 + 4t, -4 + t \rangle, \quad 0 \leq t \leq 1$$

Verification: $t = 0 \Rightarrow P_0(-1, -4, -4) \checkmark$, $t = 1 \Rightarrow P_1(2, 0, -3) \checkmark$.

🔗 P2 · Line with a Zero Direction Component

Find equations of the line through $P_0(3, -2, 5)$ with $\vec{v} = \langle 4, 0, -7 \rangle$.

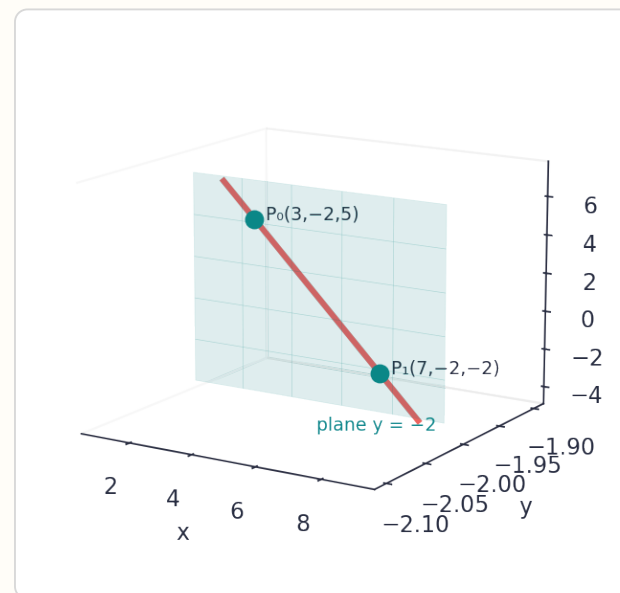
Parametric: $x = 3 + 4t$, $y = -2$, $z = 5 - 7t$ ($t \in \mathbb{R}$).

Since $b = 0$, y is strictly constant across the entire line.

Symmetric Form: Division by zero is forbidden, so state the valid fractions and isolate y :

$$\frac{x-3}{4} = \frac{z-5}{-7}, \quad y = -2$$

Geometry: The entire line is embedded in the plane $y = -2$.



Trapped in plane $y = -2$.

P3 · Classifying a Shifted Quadric by Completing the Square

Classify the surface

$$x^2 - y^2 + 4z^2 - 2x - 4y + 8z + 1 = 0$$

Step 1: Group and complete squares per variable:

$$x^2 - 2x = (x - 1)^2 - 1$$

$$-y^2 - 4y = -(y^2 + 4y) = -[(y + 2)^2 - 4] = -(y + 2)^2 + 4$$

$$4z^2 + 8z = 4(z^2 + 2z) = 4[(z + 1)^2 - 1] = 4(z + 1)^2 - 4$$

Step 2: Substitute and combine constants:

$$(x - 1)^2 - 1 - (y + 2)^2 + 4 + 4(z + 1)^2 - 4 + 1 = 0$$

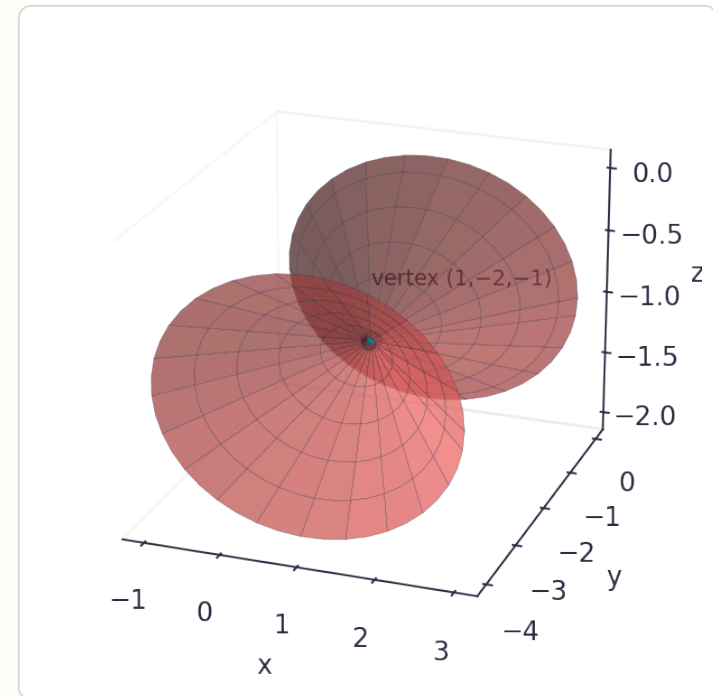
$$(x - 1)^2 - (y + 2)^2 + 4(z + 1)^2 = 0$$

Step 3: Isolate and classify:

$$(y + 2)^2 = (x - 1)^2 + 4(z + 1)^2$$

Right side degree 2, RHS = 0 → **Elliptic Cone**.

Vertex: $(1, -2, -1)$. **Axis:** parallel to y -axis.



Shifted cone vertex $(1, -2, -1)$.

Stewart 9e §12.5 1-5, 13, 17, 23-39

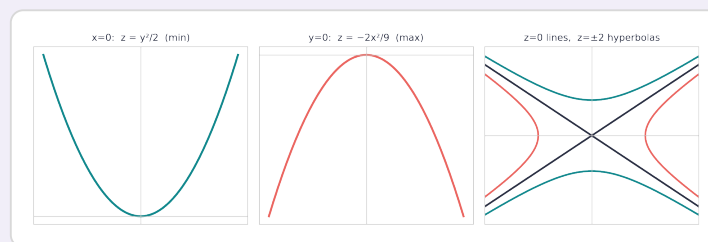
Stewart 9e §12.6 1-7, 23-30, 41, 55

2 · Trace Analysis Master Problem

🔗 P4 · Complete Trace Analysis & Surface Identification — $\frac{z}{2} = \frac{y^2}{4} - \frac{x^2}{9}$

Slice Plane	Trace Equation	Geometric Curve	Physical Behavior at Origin
$x = 0$ (in yz)	$z = \frac{y^2}{2}$	Upward parabola (V at origin)	Along y -axis, origin is a local minimum
$y = 0$ (in xz)	$z = -2\frac{x^2}{9}$	Downward parabola (V at origin)	Along x -axis, origin is a local maximum
$z = 2$ (above origin)	$\frac{y^2}{4} - \frac{x^2}{9} = 1$	Hyperbola opening along y	Cross-sections open north/south
$z = -2$ (below origin)	$\frac{x^2}{9} - \frac{y^2}{4} = 1$	Hyperbola opening along x	Cross-sections flip to east/west
$z = 0$ (in xy plane)	$y = \pm(\frac{2}{3})x$	Two intersecting lines through origin	Asymptotic boundary between level curves

Classification: Because upward curvature along y meets downward curvature along x , the origin $(0,0,0)$ is a **saddle point**. The surface is a **hyperbolic paraboloid** opening along z .



Trace planes flipping axes across $z = 0$.

3 · Test Yourself (Active Recall Worksheet)

Designed for iPad Noteful handwriting with Apple Pencil. Complete before checking the Answer Key.

Q1

A line passes through $A(2, -1, 6)$ and $B(2, 5, -3)$.

- (a)** Find the direction vector $\vec{v}$ and write the parametric equations. **(b)** Write the symmetric equations (identify any zero direction components).
(c) Write the vector equation of the line segment from A to B and verify both endpoints.

Q2

Classify each surface and state its axis of symmetry or opening direction:

(a) $9x^2 - y^2 + 4z^2 = 1$

(c) $4x^2 + y^2 - z^2 = 0$

(b) $x = 2y^2 + 3z^2$

(d) $z = x^2 - 4y^2$

Q3

Consider the equation $x^2 + 4y^2 - 6x + 8y - z + 9 = 0$.

- (a)** Complete the square and express the equation in standard form.
- (b)** Identify the surface type and state the coordinates of its vertex.
- (c)** Write the Maple command to plot this surface centered around its vertex.

Solutions & Answer Key

Check answers only after attempting. Use for self-grading in Noteful.

A1

(a) $\vec{v} = \overrightarrow{AB} = \langle 2 - 2, 5 - (-1), -3 - 6 \rangle = \langle 0, 6, -9 \rangle$.

Parametric: $x = 2, \quad y = -1 + 6t, \quad z = 6 - 9t, \quad t \in \mathbb{R}$.

(b) $a = 0$, so division by zero is illegal:

$$\frac{y+1}{6} = \frac{z-6}{-9}, \quad x = 2$$

The line lives in the vertical plane $x = 2$.

(c) $\vec{r}(t) = \langle 2, -1 + 6t, 6 - 9t \rangle, \quad 0 \leq t \leq 1$.

Check: $t = 0 \Rightarrow (2, -1, 6) = A \checkmark; t = 1 \Rightarrow (2, 5, -3) = B \checkmark$.

A2

(a) One minus sign, $\text{RHS} = 1 \rightarrow$ **Hyperboloid of one sheet**, axis = y .

(b) x is linear, two quad terms same sign $\rightarrow$ **Elliptic Paraboloid**, opens along $+x$.

(c) All degree 2, $\text{RHS} = 0 \rightarrow$ **Elliptic Cone**, axis = z .

(d) z linear, opposite quadratic signs $\rightarrow$ **Hyperbolic Paraboloid** (saddle).

A3

(a) Group terms and complete squares:

$$x^2 - 6x = (x - 3)^2 - 9$$

$$4y^2 + 8y = 4(y^2 + 2y) = 4[(y + 1)^2 - 1] = 4(y + 1)^2 - 4$$

$$(x - 3)^2 - 9 + 4(y + 1)^2 - 4 - z + 9 = 0$$

$$z + 4 = (x - 3)^2 + 4(y + 1)^2$$

(b) **Elliptic Paraboloid**, opening along $+z$, vertex $(3, -1, -4)$.

(c) Maple command:

```
with(plots):
plot3d((x-3)^2 + 4*(y+1)^2 - 4,
       x=0..6, y=-4..2,
       style=patchcontour, axes=boxed);
```